



## API ARCHITECTURE

# REST

VS

# GRAPHQL

Understanding the two most popular API architectures and when to use each

**MULTIPLE ENDPOINTS**

**SINGLE ENDPOINT**



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# What is REST?

**REST** (Representational State Transfer) is a way of building APIs where you have **different URLs** for different data, and each URL gives you **fixed data**.



## Multiple Endpoints

Each resource has its own URL



## HTTP Methods

GET, POST, PUT, DELETE



## Fixed Response

Server decides what data to send



## Stateless

Each request is independent



**Think of REST like a restaurant menu:** You go to /pizza for pizza, /pasta for pasta. Each item has a fixed serving. You can't customize what you get!



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# REST API in Action

Just like ordering from a fixed menu. You pick an item (endpoint), and the waiter brings exactly what's listed. No substitutions!

REST API Request: Ordering

Can I have a /pizza from the menu?

**MENU**  
(API Endpoints)

- /pizza ☐
- /pasta ☒
- /kadai paneer ☒
- /chhole bature ☐
- /momos ☐

Client requests specific resources via endpoints.

REST API Response: Delivering (Data format)

Here is your /pizza because you ordered pizza

REST API Request: Ordering

Can I have a /pasta from the menu?

**MENU**  
(API Endpoints)

- /pizza ☐
- /pasta ☒
- /kadai paneer ☒
- /chhole bature ☐
- /momos ☐

Client requests specific resources via endpoints.

REST API Response: Delivering (Data format)

Here is your /pasta because you ordered pasta.



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# What is GraphQL?

**GraphQL** is a query language for APIs where you have **one endpoint** and you ask for **exactly what you need**. No more, no less!



## Single Endpoint

Everything through /graphql



## Flexible Queries

Ask only what you need



## Combine Requests

Multiple queries in one call



## Typed Schema

Clear data structure defined



**Think of GraphQL like talking to a chef:** "Give me just the pizza toppings" or "Only the paneer from kadai paneer". You have full control over what you get!



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# GraphQL API in Action

Like talking directly to the chef! You specify exactly what ingredients you want, and the chef prepares only that. No waste, no extra!

GRAPQL API Request: Asking Specific Data

Can I get only pizza toppings, and only paneer from kadai paneer

ITEMS

- pizza
- pizza { toppings, crust, crust, sauce }
- pasta { type, sauce }
- pasta { sauce }
- kadai paneer { spices }

Client requests specific fields from the resources.

GRAPQL API Response: Receiving Specific Data

Here's your only pizza toppings and only paneer from kadai paneer

```
query {  
  pizza {  
    toppings {  
      mushrooms  
    }  
  }  
  kadai_paneer {  
    cubes  
  }  
}
```

JSON Data

```
{  
  "data": {  
    "pizza": {  
      "toppings": [ "mushrooms" ]  
    },  
    "paneer": {  
      "cubes": true  
    }  
  }  
}
```

Server provides only the requested data fields.

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# Request & Response Example

REST API

Client Sends Request:

GET /users/10

Server Sends Response (Fixed):

```
{  "id": 10,  "name": "Aisha",  "email": "aisha@example.com",  "age": 22,  "address": "Delhi"}
```

⚠️ Even if you need only name, you get everything!

GraphQL API

Client Sends Query:

```
{  user(id: 10) {    name    email  }}
```

Server Sends Response (Exact):

```
{  "data": {    "user": {      "name": "Aisha",      "email": "aisha@example.com"    }  }}
```

✅ You asked for name & email, that's all you get!



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# Key Differences

| Feature                                                                                                      | REST                                       | GraphQL                                  |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------|
|  <b>Endpoints</b>            | Multiple URLs<br>/users, /posts, /comments | Single URL<br>/graphql                   |
|  <b>Data Fetching</b>        | Fixed response<br>(Over-fetching)          | Flexible response<br>(Ask what you need) |
|  <b>Multiple Resources</b> | Multiple requests<br>needed                | Single request<br>can fetch all          |
|  <b>Learning Curve</b>     | Easy to learn<br>and implement             | Requires learning<br>query syntax        |
|  <b>Best For</b>           | Simple apps,<br>CRUD operations            | Complex apps,<br>mobile apps             |



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